

Prototyping Assignment

AI-Powered RFQ → Vendor Evaluation → Award Decision System

Important Note

This should be a **working AI prototype**.

- No static dashboards
 - No hardcoded outputs
 - AI must actively process inputs and generate results
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Goal (in simple terms)

Build a working system that:

- 👉 Takes an RFQ
- 👉 Sends structured + AI-generated questionnaires to vendors
- 👉 Accepts messy vendor quotation documents
- 👉 Extracts and compares them
- 👉 Helps a buyer decide **who to award and why**

Why This Assignment Exists

In real procurement workflows, especially for categories like marketing services, the process looks like this:

- A buyer creates an RFQ with scope, timelines, and line items
- Multiple vendors respond, but not in a standard format
- Each vendor sends **their own style of documents** (PPTs, PDFs, Excel sheets)
- Information is scattered, inconsistent, and often incomplete
- The buyer spends hours manually:
 - reading documents
 - comparing quotes
 - aligning scope
 - making trade-offs

👉 This process is slow, error-prone, and hard to scale

Where AI Comes In

AI can help:

- structure unstructured documents
- extract and normalize key data
- highlight differences and risks
- support better, faster decision-making

But in reality:

👉 **It's not just about generating text**

👉 It's about building a **reliable system that works end-to-end**



How This Relates to the Role

As a Senior Applied AI Engineer, your day-to-day work will involve:

- Taking ambiguous, real-world problems like this
- Designing AI-powered systems end-to-end
- Deciding:
 - where to use AI
 - where to use rules
 - how to make outputs reliable
- Building fast prototypes that can evolve into product features
- Working closely with product and business teams to deliver real value

👉 This assignment simulates that exact responsibility



What We Want to See

We are not just looking for:

- a UI, or a single AI prompt

We are looking for:

- how you think about the problem
 - how you structure the system
 - how you make outputs useful and trustworthy
 - how you balance simplicity and effectiveness
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One Important Note

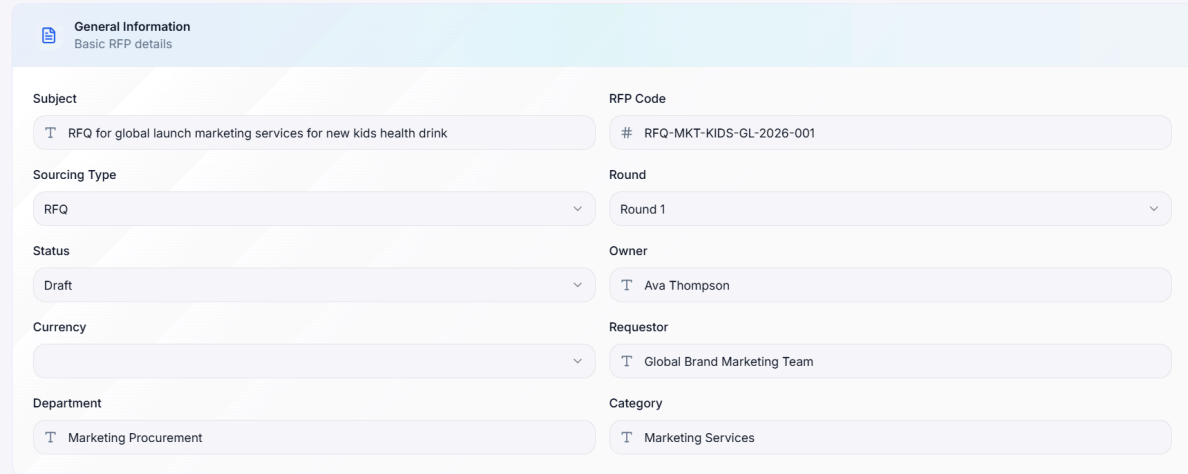
If anything is unclear, **please reach out**. We prefer thoughtful questions over incorrect assumptions.

End-to-End Flow (what you need to build)

Page 1: Step A : RFQ + Questionnaire Generation

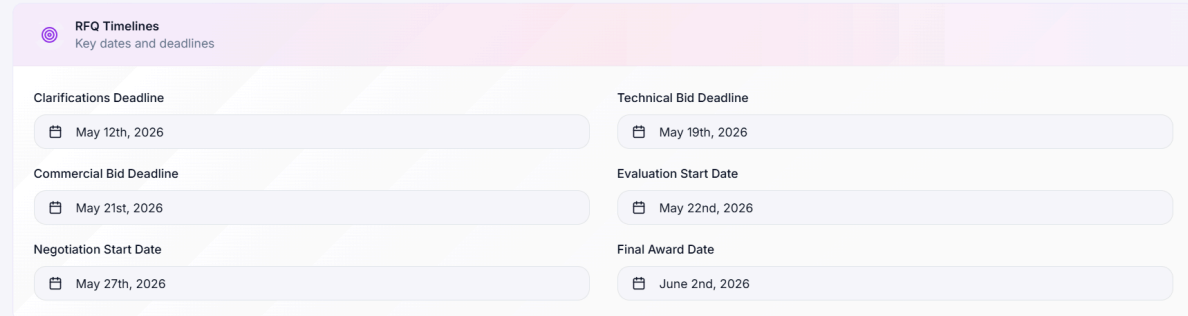
A sample RFQ has these components - General Information Section, Timelines, Item Requests, Scope of Work and Questionnaire

Develop this page using the following screenshots



General Information
Basic RFP details

Subject T RFQ for global launch marketing services for new kids health drink	RFP Code # RFQ-MKT-KIDS-GL-2026-001
Sourcing Type RFQ	Round Round 1
Status Draft	Owner T Ava Thompson
Currency 	Requestor T Global Brand Marketing Team
Department T Marketing Procurement	Category T Marketing Services



RFQ Timelines
Key dates and deadlines

Clarifications Deadline May 12th, 2026	Technical Bid Deadline May 19th, 2026
Commercial Bid Deadline May 21st, 2026	Evaluation Start Date May 22nd, 2026
Negotiation Start Date May 27th, 2026	Final Award Date June 2nd, 2026

* Have all the 8 items in your RFQ page.

#	Product Name	Category	Description	HSN/SAC	UOM
1	Strategy & Creative Development	Services	Launch strategy, audience segmentation, messaging framework, creative territory development and master campaign toolkit for a new kids health drink.	8471XX	Lot
2	TVC Development	Services	Development of flagship TV commercial concept and all pre-production creative required for global launch approvals.	8471XX	Lot
3	TVC Production	Services	End-to-end TVC shoot, post-production and delivery of master film and cutdowns for paid media use.	8471XX	Lot
4	Social Organic Content	Services	Organic social content strategy, monthly content calendar and asset adaptation for Meta, YouTube and TikTok.	8471XX	Lot
5	Social Paid Media Planning	Services	Paid social and video media planning across Meta, YouTube and TikTok for global launch period.	8471XX	Lot
6	Social Paid Media Buying & Optimization	Services	Campaign trafficking, paid media activation, optimization and performance reporting across Meta, YouTube and TikTok.	8471XX	Lot
7	Kids Advertising & Claims Compliance Review	Services	Legal and regulatory review of kids advertising content and product-related claims across launch assets and scripts.	8471XX	Lot
8	Launch Program Management	Services	Program management, stakeholder coordination, asset trafficking and master launch governance across all workstreams.	8471XX	Lot

Scope of Work

Statement of Work defining scope, objectives, and deliverables

AI Assist

H1H2H3BIL

The selected vendor(s) will support the global launch of a new kids health drink across strategy, creative development, film production, social activation and compliance review. The scope is intentionally structured to allow either a single integrated award or a multi-vendor award by workstream.

1. Strategy & Creative Development: Develop launch strategy, audience understanding, positioning, campaign idea, messaging architecture and master visual direction suitable for parents, families and child-safe communication environments. Deliverables include strategic narrative, creative territory options, key visuals, copy routes and channel adaptation guidance.

2. TVC Development: Create the flagship film idea and all pre-production materials including scripts, storyboard, animatic, packshot plan, supers and production approach. Concepts must be suitable for a kids/family product and avoid unsupported product or health claims.

3. TVC Production: Execute end-to-end film production, including pre-production, casting, shoot, post-production, finishing and final delivery of master and cutdown assets. Vendors must account for child talent safety, usage rights, market-ready master files and trafficking-ready outputs.

4. Social Organic: Build a launch-phase organic social content plan for Meta, YouTube and TikTok. Deliver content calendars, platform-adapted assets, copy and publishing guidance aligned to the campaign idea and local sensitivities for child-directed communications.

5. Social Paid: Provide paid media planning and/or activation services across Meta, YouTube and TikTok. Plans should prioritize age-appropriate placements, parent/co-viewing audiences, launch objectives, reporting cadence and optimization logic. If media buying is proposed, agency fees must be clearly separated from pass-through media spend.

6. Kids Advertising & Claims Compliance Review: Review scripts, visuals, videos, copy and product-related claims against applicable advertising standards, child-marketing restrictions, nutrition/health claim guardrails and platform policies. Compliance notes should be practical, market-aware and timed to avoid launch delay.

7. Program Management: Manage project timelines, approvals, stakeholder coordination, version control and final asset delivery across all awarded workstreams.

All vendors must clearly specify what is included, excluded, dependent on client inputs and subject to third-party pass-through costs. Commercial submissions should map directly to the 8 RFQ line items and maintain the same line-item structure in the response.

Show Less

Your system should:

1. Read the other RFQ component’s data (scope, line items, timelines, etc.)
2. A button exists on the page clicking on which hits an AI agent to generate **vendor questionnaires (Hint: There can be multiple types of questionnaires)**

Show the questionnaire section on the UI too.

Example

If RFQ says:

“TVC production with child safety compliance”

Your system may generate:

- What is your experience with child talent compliance?
 - Share 3 relevant past campaigns
 - What safety processes do you follow?
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Page 2: Step B : Vendor Responses (Messy Documents)

Assume:

- RFQ is sent to **5 vendors**
- Each vendor uploads **1 document** (any format)

You must support:

- PDF
- PPT
- Excel
- Word

👉 Build a simple upload page for this

⚠️ **Make documents realistically messy**

Your system should handle real-world chaos.

Include at least 5 of these in your test data:

1. Pricing hidden in **tables inside PDFs**
2. Important info present as **images in PPT**
3. Vendors giving **partial quotes (missing fields)**
4. Very long documents (20–50 pages)
5. Different formats for same thing
 - “Total Cost” vs “Grand Total” vs “Project Fee”
6. Mixed currencies or unclear units
7. Commercials split across multiple pages

👉 You can create dummy documents yourself

Page 3: Step C : Extraction + Normalization

Your system should:

1. Extract:
 - Pricing
 - Commercial terms
 - scope coverage
 - delivery timelines
 - key terms
2. Normalize:
 - convert everything into same structure
 - align line items with RFQ
 - standardize naming

👉 This is a **critical step** : don't skip structure

Page 4: Step D : Technical Analysis Page

Once all the quotations are received and normalized and stored in a place, System should generate live analysis on click of a button (to be shown on a dashboard):

- Which vendors meet requirements
- Where vendors differ
- Missing responses
- Strengths / weaknesses
- ... and think of ways which technical analysis showcase would look good

👉 Must include **evidence (from documents)**

Page 5: Step E : Commercial Comparison Dashboard

Once all the quotations are received and normalized and stored in a place, System should generate live analysis on click of a button (to be shown on a dashboard):

- cost comparison
 - cost breakdown
 - anomalies (too high / too low)
 - missing pricing
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Page 6: Step F : Award Recommendation (MOST IMPORTANT)

Your system should:

1. Suggest:
 - best vendor OR
 - split award (e.g., Vendor A for production, Vendor B for media)
2. Run **multiple scenarios**, for example:
 - lowest cost
 - fastest delivery
 - best compliance
3. Explain:
 - why this recommendation
 - trade-offs
 - confidence / risks

Important for Award Recommendation

Your system should NOT:

- always pick lowest cost blindly

Your system SHOULD:

- consider multiple factors (cost, compliance, scope, etc.)
- allow trade-offs
- explain reasoning clearly

 Even a simple scoring model is acceptable, but it must be **logical and explainable**



What We Are Evaluating

1. End-to-End Thinking

Did you build a system or just isolated pieces?

2. Product Thinking

Is the flow intuitive?, Can a buyer trust the output?

3. AI Usage (IMPORTANT)

We are NOT evaluating prompting only. We care about how you structure the system, where you use AI vs rules, how you validate outputs

4. Handling Messy Data

Real procurement data is not clean.

5. Explainability

Every insight should be traceable.

6. UX (a Polished UI would be a good addition)

- clear flow, easy to understand outputs, logical structure
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7. AI Generated Outputs are Grounded

All AI-generated insights, data, etc -> must be backed by source data from vendor documents, reference where the information came from (on the UI)

Deliverables

TIMEFRAME: You are expected to submit this assignment within 5 days of receiving it.

1. Working Prototype

- Web app link
 - Should be runnable
 - Include sample vendor documents
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2. Demo Video (5 mins)

Explain:

- system flow
 - key decisions
 - live demo
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3. Transcript of Video

4. 1–2 Page Document

Cover:

- problem you solved
 - assumptions
 - system design (high level)
 - key features
 - what you would improve
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5. (Optional but HIGHLY valuable)

- architecture diagram (high level)
 - evaluation approach
 - logs / traces
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Guidance

You are encouraged to use AI tools.

But we are evaluating:

👉 how you **design systems**, not just generate outputs



What to Avoid

- Dummy filled clickable dashboards
 - Easy documents from vendor side
 - No evidence for AI generated outputs
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What Strong Submissions Will Show

- clear pipeline (extract → normalize → analyze → decide)
 - mix of AI + logic
 - grounded outputs
 - thoughtful UX, with a good looking UI is a cherry on top
 - meaningful recommendations
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Final Note

This assignment is intentionally open-ended.

There is no “one correct answer.”

We are looking for:

👉 how you turn a messy real-world problem into a usable system
